

# Ivan Brillo

AI Researcher · MSc in AI & Data Engineering · Robotics Honors Student

✉ brilloivan2@gmail.com

📍 Pisa, Italy

🌐 ivanbrillo.github.io

🔗 ivanbrillo

🌐 ivan-brillo

## Summary

AI researcher working on machine learning for science and healthcare — surrogate modelling and Bayesian optimization, deep learning for physiological signals (ECG, EEG), and explainable AI — with a background in embedded systems.

## Education

|                                                                                                                                                                                                                                                                                                                                                                   |                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| <b>Sant'Anna School of Advanced Studies</b> , Robotics <ul style="list-style-type: none"> <li>Awarded 2-Year Merit Full Scholarship (top 2% of applicants).</li> </ul>                                                                                                                                                                                            | Pisa, Italy<br>Sept 2024 – present       |
| <b>University of Pisa</b> , Artificial Intelligence and Data Engineering <ul style="list-style-type: none"> <li>Thesis: "Development and Evaluation of a Domain-Adversarial Transformer for Subject-Invariant Physiological Signals Analysis". Supervisor M. Cimino.</li> </ul>                                                                                   | Pisa, Italy<br>Sept 2024 – July 2026     |
| <b>Budapest University of Technology and Economics</b> , Challenge-based human-inspired computer vision                                                                                                                                                                                                                                                           | Budapest, Hungary<br>Dec 2024 – Dec 2024 |
| <b>University of Padua</b> , Computer Engineering <ul style="list-style-type: none"> <li>DECAMP Course in Applied Web Application Security (Metropolia UAS, Finland).</li> <li>Thesis: "A deep learning approach for the analysis of calcium signal in characterizing the biological response of plants to external stimuli". Supervisor E. Menegatti.</li> </ul> | Padua, Italy<br>Oct 2021 – July 2024     |
| <b>Scientific High School G.B. Ferrari</b> , Applied Sciences                                                                                                                                                                                                                                                                                                     | Este, Italy<br>Sept 2016 – July 2021     |

## Experience

|                                                                                                                                                                                                                                                                                                        |                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>SLAC National Accelerator Laboratory (Stanford University)</b> , Visiting Scientist — Vernieri Group<br>Developing an agentic AI system to automate the optimization and design of Future Circular Collider (FCC) particle detectors. Supervisor C. Vernieri.                                       | Menlo Park, CA, USA<br>July 2026 – Sept 2026<br>3 months |
| <b>Mila — Quebec Artificial Intelligence Institute</b> , Visiting Researcher — sciNeurotech Lab (Polytechnique Montréal)<br>Developed CNN surrogates to replace computationally expensive online Gaussian Process updates. Funded by a Mitacs Globalink Research Scholarship. Supervisor M. Bonizzato. | Montréal, Canada<br>Mar 2026 – June 2026<br>4 months     |
| <b>Research Center "E. Piaggio"</b> , Student Researcher — MLPI Lab<br>Framing R-peak prediction from EEG as a segmentation problem via distance-to-peak estimation, evaluating multi-scale CNNs and autoregressive models with a focus on domain generalization. Supervisor M. Cimino.                | Pisa, Italy<br>Oct 2025 – July 2026<br>10 months         |
| <b>Monasterio Foundation</b> , Student Researcher — PERCRO Lab<br>Cardiac amyloidosis classification from ECGs using Transformers, with interpretability via Grad-CAM and Attention Rollout. Supervisor C.A. Avizzano.                                                                                 | Pisa, Italy<br>Sept 2025 – Jan 2026<br>5 months          |

|                                                                                                                                                                                                                                                                                                                                                                                 |                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| <b>Sant'Anna School of Advanced Studies</b> , Student Researcher — NeuroRobotic Touch Lab and TeCIP Institute <ul style="list-style-type: none"> <li>Explainable classification of tumor-mimicking materials from force/position data. Supervisor C.M. Oddo.</li> <li>Efficient federated learning via dynamic training-precision adjustment. Supervisor E. Paolini.</li> </ul> | Pisa, Italy<br>Feb 2025 – Oct 2025<br>9 months          |
| <b>UP caeli via</b> , Computer Engineer — Part Time (24h/week)<br>Software integration for drones (QGC and ArduPilot) using C++ and Qt/QML (GeoFencing, RTK and RTSP).                                                                                                                                                                                                          | Padua, Italy<br>Mar 2024 – July 2024<br>5 months        |
| <b>RFX Consortium and CNR</b> , Internship — Nuclear Fusion and Plasma Research<br>Investigated autoencoders and VAEs for temporal analysis of plant stress responses after irrigation with plasma-activated water (PAW). Supervisor V. Antoni.                                                                                                                                 | Padua, Italy<br>Nov 2023 – Feb 2024<br>4 months         |
| <b>Formula Student Team Project</b> , Embedded Software Developer (+10h/week)<br>Member of a 50-person university racing team, developing embedded systems (ATmega32/64, TI F2837, Arduino) and C control algorithms for power management, torque vectoring, and regenerative braking.                                                                                          | Padua, Italy<br>Oct 2022 – Aug 2024<br>1 year 11 months |

## Publications

---

|                                                                                                                                                                                                                                                                                                                                                                                                           |           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>AI-Driven Optimization of Detector Design for the FCC-ee</b><br>Presented research on multifidelity Bayesian optimization and differentiable simulation surrogates for efficient design optimization for the proposed FCC-ee at CERN.<br><b>Ivan Brillo</b><br><a href="https://indico.slac.stanford.edu/event/10213/contributions/15916">indico.slac.stanford.edu/event/10213/contributions/15916</a> | Sept 2026 |
| <b>ResCTrans: an Explainable Amyloidosis Classifier using Temporal Transformers from ECG Signals</b><br><b>Ivan Brillo</b> , et al.                                                                                                                                                                                                                                                                       | July 2026 |

## Awards

---

|                                                                                                                                                                        |           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>Zegna Scholarship</b><br>Scholarship awarded for academic excellence and merit.<br>Zegna                                                                            | June 2026 |
| <b>Mitacs Globalink Research Scholarship</b><br>Scholarship supporting a visiting research stay at Mila.<br>Mitacs                                                     | Mar 2026  |
| <b>"Mille e una Lode" Merit Awards</b><br>Scholarships awarded to the top 3% of students, received in 2022, 2023, and 2024.<br>University of Padua                     | July 2024 |
| <b>INPS Scholarships</b><br>National scholarships in recognition of academic excellence, received annually 2018–2024.<br>Italian National Institute of Social Security | July 2024 |
| <b>National Bronze Medal, Physics Olympiad</b><br>Association for Teaching Physics                                                                                     | May 2021  |
| <b>National First Place, Chemistry Olympiad</b><br>Italian Chemistry Society                                                                                           | June 2020 |

## Memberships

---

**LeadTheFuture — Mentee (Oct. 2025–present):** Selective STEM mentorship program, acceptance rate < 15%.

**ISSNAF — Mentee (Dec. 2025–present):** Selective mentorship for research opportunities in the US.

## Skills

---

**Machine Learning**

**Programming**

**Embedded & Robotics**

**Data & Tools**

## Languages

---

**Italian**

Native

**English**

C1 (TOEFL iBT 104/120)